MIL-E-24403/1D(SH)
14 November 1985
SUPERSEDING
MIL-E-24403/1C(SH)
30 August 1983
(See 6.3)

MILITARY SPECIFICATION

ELECTRODES - WELDING, FLUX CORED, ORDINARY STRENGTH AND LOW ALLOY STEEL

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ordinary strength and low alloy steel flux cored welding electrodes for use with the flux cored arc welding process with or without shielding gas.

1.2 Classification. Electrodes shall be of the types specified in table I, and of the forms and sizes in accordance with MIL-E-24403. In addition to the sizes specified in MIL-E-24403, the 0.035-inch size may be used for MIL-71T-1, MIL-81T1-Ni1 and MIL-81T1-Ni2 electrodes, and the 0.068- and 0.072-inch sizes may be used for self-shielded electrodes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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TABLE I. Type designation and chemical composition of deposited weld metal.[1]

MIL-type	Chemical composition (percent)[2]										
	Carbon	Manganese	Silicon	Phos-phorus	Sulfur	Nickel	Chro-mium	Molyb-denum	Vanadium	Copper	Aluminum[3]
[4]MIL-70T-1 [4]MIL-71T-1 [4]MIL-70T-5 [4]MIL-71T-1A	0.12	0.50 to 1.75	0.90	0.030	0.030	0.50	0.20	0.30	0.05	0.20	----
[5]MIL-70T-6 [5]MIL-70T-8 [5]MIL-71T-8	0.12	0.70 to 1.80	0.40	0.030	0.030	0.30	0.20	0.30	0.05	0.20	1.10
[5]MIL-71T-8A	0.25	0.20 to 1.80	0.60	0.030	0.030	0.30	0.20	0.30	0.05	0.20	1.10
[5]MIL-71T8-K6	0.15	0.50 to 1.50	0.40	0.030	0.030	0.40 to 1.10	0.20	0.30	0.05	0.20	1.10
[5]MIL-71T8-Ni1	0.12	0.70 to 1.80	0.40	0.030	0.030	0.80 to 1.40	0.20	0.30	0.05	0.20	1.10
[4]MIL-80T1-Ni1 [4]MIL-81T1-Ni1	0.12	0.50 to 1.80	0.80	0.030	0.030	0.80 to 1.10	0.15	0.35	0.05	0.20	----
[4]MIL-80T1-Ni2 [4]MIL-81T1-Ni2	0.12	0.50 to 1.80	0.80	0.030	0.030	1.75 to 2.75	----	0.30	0.05	0.20	----

[1] Sizes 0.035, 0.045, 0.052, 0.0625, 0.068, 0.072, 5/64, 3/32, 7/64, 0.120, 1/8, 5/32.
[2] Single values are maximum percentages.
[3] For self-shielded electrodes only.
[4] These electrodes shall have "C" or "M" suffix. Electrodes qualification tested and quality conformance tested using carbon dioxide shielding gas shall have a "C" suffix, and electrodes tested using 75 percent argon and 25 percent carbon dioxide shielding gas mixture shall have an "M" suffix (for example 70T-1C or 70T-1M).
[5] These self-shielding electrodes shall have an "S" suffix.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standard. The following specifications and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-C-101 - Carbon Dioxide (CO_2): Technical and U.S.P.

MILITARY

MIL-A-18455 - Argon, Technical.
MIL-S-22698 - Steel Plate and Shapes, Weldable Ordinary Strength and Higher Strength: Hull Structural.
MIL-E-24403 - Electrodes - Welding, Flux Cored, General Specification for.

STANDARD

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.

2.1.2 Government publication. The following Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 131 - Standard Specification for Structural Steel for Ships.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-24403 and as specified herein.

3.2 Finish. Copper or other suitable coatings may be used.

3.3 Chemical composition. Chemical composition of the deposited weld metal shall be as specified in table I.

3.4 Mechanical properties. The mechanical properties of deposited weld metal shall be as specified in table II.

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TABLE II. Mechanical property requirements.

MIL-type	Ultimate tensile strength minimum lb/in² as-welded[1]	Yield strength at 0.2 percent offset, minimum lb/in² [1]		Elongation in 2 inches, minimum percent[1]		Charpy V-notch impact[2]	
		As-welded	Stress-relieved[4]	As-welded	Stress-relieved[4]	Energy ft-lb minimum average as-welded and stress-relieved[4]	Temperature °F[3]
MIL-70T-6 MIL-70T-1 MIL-71T-1	70,000	60,000	55,000	22	22	20 30	0 30
MIL-71T-1A MIL-70T-5 MIL-70T-8 MIL-71T-8 MIL-71T-8A MIL-71T8-K6 MIL-71T8-Ni1	70,000	60,000 to 75,000	55,000	22	22	20 35	minus 20 0
[5]MIL-7XT-1-HY [5]MIL-7XT-1-HZ	70,000	60,000 to 75,000	55,000	22	22	20 35	minus 20 0
[5]MIL-7XT-X-HY [5]MIL-7XT-X-HZ [5]MIL-7XTX-X-HY [5]MIL-7XTX-X-HZ	70,000	60,000 to 75,000	55,000	22	22	20 35	minus 20 0
[5]MIL-8XTX-X-HY [5]MIL-8XTX-X-HZ	80,000	65,000 to 85,000	60,000	22	22	30 40	minus 20 0
MIL-80T1-Ni1 MIL-81T1-Ni1 MIL-80T1-Ni2 MIL-81T1-Ni2	80,000	65,000 to 85,000	60,000	22	22	20 35	minus 20 0

See footnotes at top of next page.

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[1] Mechanical properties as determined from an all-weld metal tension test specimen.
[2] For each testing temperature, the average value of five tests shall be greater than the minimum average value specified. No two specimens shall have values below the minimum average specified. One specimen can have a value of 10 foot-pounds (ft-lb) below the minimum average specified.
[3] Temperature tolerance, plus or minus 3 degrees Fahrenheit (°F).
[4] Stress relieved procedure shall be in accordance with footnote 5 of figure 1.
[5] Electrodes qualification tested for conformance to these properties and hydrogen as specified (see 3.6) for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel shall have a suffix of "HY" or "HZ" added to the type designation (see 6.1.1.7).

3.5 Soundness. Welds shall meet the standards for radiographic inspection in accordance with class I of NAVSEA 0900-LP-003-9000.

3.6 Hydrogen. Hydrogen levels determined in accordance with MIL-E-24403 shall not exceed the following limits:

(a) For weld metal deposited with type MIL-70T-1, MIL-71T-1, MIL-70T-5, MIL-70T-6, MIL-70T-8, MIL-71T-8, MIL-71T-8A, MIL-71T8-K6, and MIL-71T8-Ni1 electrodes - maximum average value of 15.0 milliliters (mL) per 100 grams of weld metal and a maximum single value of 17.0 mL per 100 grams of weld metal.
(b) For weld metal deposited with type MIL-80T1-Ni1, MIL-81T1-Ni1, MIL-80T1-Ni2, and MIL-81T1-Ni2 electrodes - maximum average value of 10.0 mL per 100 grams of weld metal and a maximum single value of 12.0 mL per 100 grams of weld metal.
(c) For weld metal deposited with type MIL-7XT-1-HY, MIL-7XT-X-HY, MIL-7XTX-X-HY, and MIL-8XTX-X-HY electrodes - maximum average value of 5.0 mL per 100 grams of weld metal and a maximum single value of 5.8 mL per 100 grams of weld metal.
(d) For weld metal deposited with type MIL-7XT-1-HZ, MIL-7XT-X-HZ, MIL-7XTX-X-HZ, and MIL-8XTX-X-HZ electrodes - maximum average value of 2.00 mL per 100 grams of weld metal and a maximum single value of 2.30 mL per 100 grams of weld metal.

3.7 Alloy identity. For lots specified in MIL-E-24403, each end of the rod or strip to be spliced during processing shall be sampled at the splicing station for alloy identity (see 4.4), except when splicing is done to repair a break without removing the strip from the process line. The sample shall be tested for alloy identity prior to shipment of the lot.

3.7.1 Both ends of each coil rolled to final thickness shall be tested for alloy identity before further processing into electrodes (see 4.4).

3.7.2 When specified (see 6.2), each electrode shall be tested for alloy identity in accordance with 4.4 after final marking. Identification marking of each spool, coil, or container of electrodes tested after final marking and meeting the acceptance criteria shall include the words "ALLOY TESTED" (see 4.4.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-24403 and as specified herein.

4.2 Qualification - required tests. Electrodes selected in accordance with MIL-E-24403 shall be used for tests specified in table III.

TABLE III. Summary if tests required for qualification.

MIL-type electrode	Test required	Test procedures	Requirements
All	Visual and dimensional	MIL-E-24403	MIL-E-24403
	Alloy identity	4.4 herein	3.7 herein
	Chemical	MIL-E-24403	Table I herein
	Hydrogen	MIL-E-24403	3.6 herein
	Soundness	[3]Note 6 to figure 1	3.5 herein
	All weld metal tension[1]	[3]MIL-E-24403	Table II herein
	Impact[2]	[3]MIL-E-24403	Table II herein[4]

[1] Includes ultimate tensile strength, yield strength, and elongation for as-welded and stress-relieved conditions.
[2] Tests shall be conducted for the as-welded and stress-relieved conditions.
[3] Welds shall be prepared in accordance with figure 1 herein.
[4] Tests shall be conducted for conformance with the impact requirements at temperatures specified in table II.

4.3 Quality conformance inspection.

4.3.1 Lot. A lot of electrodes shall be as specified in MIL-E-24403.

4.3.2 Sampling for quality conformance. Electrodes selected in accordance with MIL-E-24403 shall be used for tests specified in table IV.

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TABLE IV. Summary of tests required for quality conformance testing.

Mil-type[1] electrode	Test required	Test procedures	Requirements
All	Visual and dimensional	MIL-E-24403	MIL-E-24403
	Alloy identity	4.4 herein	3.7 herein
	Chemical	MIL-E-24403	Table I herein
	Hydrogen	MIL-E-24403	3.6 herein
	Soundness	[3]Note 6 to figure 1	3.5 herein
	All weld metal tension[1]	[3]MIL-E-24403	Table II herein
	Impact[2]	[3]MIL-E-24403	Table II herein[4]

[1] Includes ultimate tensile strength, yield strength, and elongation.
[2] Tests shall be conducted for the as-welded condition only, when specified (see 6.2).
[3] Welds shall be prepared in accordance with figure 1 herein.
[4] Tests shall be conducted for conformance with the impact requirements at the lower temperature as specified in table II for each MIL-type.

4.4 Alloy identity.

4.4.1 Procedure. The alloy identity test method shall include chemical analysis, metal sorting devices, other approved methods or a combination of methods. The test methods shall be submitted for approval to NAVSEA.

4.4.2 Rejection criteria. If the test demonstrates that the material is not of the type specified, the material shall be rejected.

4.5 Certificate of test. A certificate of quality conformance of the dimensional tolerance determination and weld tests specified in MIL-E-24403 shall be furnished to the consignee with the shipment containing actual quantitative examination and test results including the amperage employed in making the required welds for each accepted lot included in a particular shipment. The minimum data required shall be listed and prepared on the form as shown on figure 2.

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4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 Packaging shall be as specified in MIL-E-24403.

5.2 Package size and weights. The electrodes shall be packaged in packages of the sizes and weights specified in table V.

TABLE V. Unit package sizes and weights.

Form	Package size (inches)	Weight (pounds)[1]
3a	4 od	1-1/2 or 2-1/2
3b	12 od 14 od 22 od 24 od 30 od	25 or 45 50 to 60 250 300 600 or 750
3d	6-3/4 or 12 id	[2]14, 25, 50 or 60
3e	As specified (see 6.2)	As specified (see 6.2)
4	15-1/2 od 20 od 23 od	As specified (see 6.2)

[1] Tolerances on net weight shall be plus or minus 10 percent.
[2] For electrode sizes 0.068, 0.072, 5/64, and 3/32.

6. NOTES

6.1 Intended use.

6.1.1 General. This specification is intended to cover ordinary strength and low-alloy steel flux cored welding electrodes capable of depositing radiographic quality weld metal, that in the as-welded and stress-relieved condition meets the mechanical properties given herein when welded on ordinary or higher strength steels, with the flux cored arc welding process using the procedures specified in the notes to figure 1.

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6.1.1.1 MIL-70T-1C and MIL-70T-1M. Electrodes of this classification are intended to be used with gas shielding for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal fillets.

6.1.1.2 MIL-71T-1C, MIL-71T-1M, MIL-71T-1AC and MIL-71T-1AM. Electrodes of this classification are intended to be used with gas shielding for single and multiple-pass welding. Electrodes 5/64-inch diameter and smaller are used for welding in all positions.

6.1.1.3 MIL-70T-5C and MIL-70T-5M. Electrodes of this classification are intended to be used with gas shielding for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal fillets.

6.1.1.4 MIL-70T-6S. Electrodes of this classification are intended to be used with no external shielding gas for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal fillets.

6.1.1.5 MIL-70T-8S. Electrodes of this classification are intended to be used with no external shielding gas for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal fillets.

6.1.1.6 MIL-71T-8S, MIL-71T-8AS, MIL-71T8-K6S, and MIL-71T8-Ni1S. Electrodes of these classifications are intended to be used with no external shielding gas for single and multiple-pass welding. Electrodes 5/64-inch diameter and smaller are used for welding in all positions.

6.1.1.7 MIL-7XT-X-HY, MIL-7XTX-X-HY, and MIL-8XTX-X-HY. Electrodes of these classifications are intended for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel. Electrodes with a "HZ" suffix are intended for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel without preheat.

6.1.1.7.1 MIL-7XT-X-HZ, MIL-7XTX-X-HZ and MIL-8XTX-X-HZ. Electrodes of these classifications are designed for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel without preheat.

6.1.1.8 MIL-80T1-Ni1C, MIL-80T1-Ni2C, MIL-80T1-Ni1M, and MIL-80T1-Ni2M. Electrodes of these classifications are intended to be used with gas shielding for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal fillets.

6.1.1.9 MIL-81T1-Ni1C, MIL-81T1-Ni2C, MIL-81T1-Ni1M, and MIL-81T1-Ni2M. Electrodes of these classifications are intended to be used with gas shielding for single and multiple-pass welding. Electrodes 5/64-inch diameter and smaller are used for welding in all positions.

6.2 Ordering data. Acquisition documents should specify the ordering data required in MIL-E-24403 and the following:

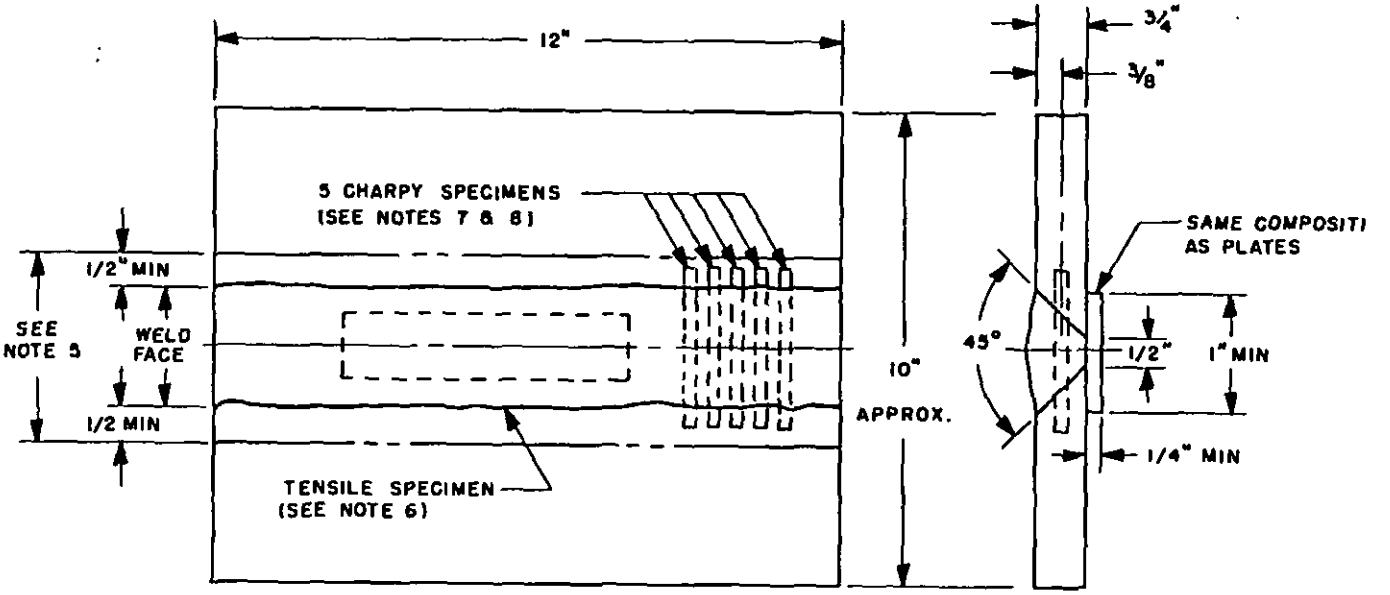
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(a) When alloy identity testing of each electrode after final marking is required (see 3.7.2).
(b) When impact tests are to be conducted in the as-welded condition (see table IV).
(c) Unit package size and weight required for electrode form 3e and unit package weight required for electrode form 4 (see table V).
(d) Whether a second weld joint for mechanical properties tests should also be made in accordance with figure 1 using a shielding gas mixture of 75 percent argon and 25 percent carbon dioxide (see notes to figure 1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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FIGURE 1. Weld joint for mechanical properties tests.

Notes to figure 1:

1. Base plate shall be in accordance with ASTM A 131, grade B or grade B steel of MIL-S-22698. One base plate assembly of the size shown is sufficient for the specimens required for quality conformance testing. Two base plate assemblies are required for qualification testing, one for the as-welded condition and the other for the stress-relieved condition. The base plate size of each assembly should be increased to provide five additional Charpy specimens required for the one additional testing temperature.
2. Assembly shall be at room temperature at the start of welding. The interpass temperature shall be controlled within 300 ± 25°F until welding is completed. Should welding be interrupted, the assembly shall be allowed to cool in still air at room temperature, and prior to resumption of welding the assembly shall be reheated to the interpass temperature (300 ± 25°F).
3. The current, polarity, welding position, and gas shielding (if required) for the various types and sizes of electrodes shall be in accordance with the following:

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Notes to figure 1: (Continued)

MIL-type	Size (inch)	Current and polarity	Welding position	Gas shielding
MIL-70T-1	All	Dc electrode positive	Flat	[1]
MIL-71T-1	5/64 and smaller	Dc electrode positive	Vertical	[1]
MIL-70T-5	All	Dc electrode positive	Flat	[1]
MIL-70T-6	All	Dc electrode positive	Flat	None
MIL-70T-8	All	Dc electrode negative	Flat	None
MIL-71T-8 MIL-71T-8A MIL-71T8-K6 MIL-71T8-Ni1	5/64 and smaller	Dc electrode negative	Vertical	None
MIL-80T1-Ni1 MIL-80T1-Ni2	All	Dc electrode positive	Flat	[1]
MIL-81T1-Ni1 MIL-81T1-Ni2	5/64 and smaller	Dc electrode positive	Vertical	[1]

[1] Shielding gas shall be carbon dioxide or a mixture of 75 percent argon and 25 percent carbon dioxide for qualification and quality conformance testing. The carbon dioxide (CO_2) shall be in accordance with BB-C-101, grade B. The argon (Ar) shall be in accordance with MIL-A-18455.

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Notes to figure 1: (Continued)

4. The pass sequence shall be in accordance with the following:

Electrode size (inch)	Passes per layer		Number of layers[1]
	Layer 1	Layer 2 to top	
0.045, 0.052, 0.0625, [3]0,068, [3]0.072, 5/64, 3/32, 7/64, 0.120, 1/8 and 5/32	1 or 2	[2]2 or 3	4 to 8

[1] Each layer shall be approximately 1/8 inch thick.
[2] Final layer may be four passes.
[3] For forms 3d, 3e, and 4 only.

5. Shielding gas, when required, shall conform to BB-C-101, grade B for carbon dioxide (CO_2) or shall conform to MIL-A-18455 for argon (Ar).
6. After completion of the weld, each assembly required shall be allowed to cool, the weld reinforcement and the backing strip shall be removed flush with the base plate on both surfaces and the weld radiographed for soundness in accordance with level 2-2T of MIL-STD-271 and 3.5 herein. The weldment shall be cut as shown resulting in a tensile coupon suitable for 0.505-inch diameter specimen and a coupon of size sufficient for five Charpy V-notch specimens for each testing temperature required. No base metal shall be removed within 1/2 inch of the edges of the faces of the weld by flame cutting.
7. When tests are required in both the as-welded and stress-relieved conditions, one tensile coupon and five impact coupons shall be stress-relieved using an electric furnace. The coupons shall be heated to 1150 ± 25°F, held for 2 hours at temperature, and allowed to furnace cool at a rate not to exceed 200°F per hour to a temperature below 500°F.
8. The 0.505-inch tensile specimen shall be machined and tested as specified in MIL-E-24403.
9. The Charpy V-notch specimens shall be machined and tested as specified in MIL-E-24403. One set of five specimens is required for each testing temperature (see table III or table IV).
10. Impact properties for the test specimens shall be obtained at test temperatures specified in table III and table IV.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or distributor _____

Address _____

Date _____

Specification MIL- _____

Type MIL- _____

Diameter _____

Customer's name _____

Customer's order no. _____

Sheathing strip heat no. _____

Lot identification no. _____

Chemical analysis

Carbon _____
Manganese _____
Silicon _____
Phosphorus _____
Sulfur _____
Nickel _____
Chromium _____
Molybdenum _____
Vanadium _____
Copper _____
Aluminum _____

Chemistry was taken from:

☐ Weld metal pad

☐ Groove weld

☐ Fractured tensile specimen

Mechanical tests

Tensile test _____
Yield strength (0.2 percent offset, lb/in^2) _____

Tensile strength (lb/in^2) _____
Elongation (percent in 2 inches) _____

Impact test

Energy ft-lb	Temperature °F (see table III,[4])
1. _____	
2. _____	
3. _____	
4. _____	
5. _____	
Average _____	

See notes at end of figure.

FIGURE 2. Certification of quality conformance tests.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS - Continued

Hydrogen analysis

1. _____ mL per 100 grams or weld metal
2. _____
3. _____
4. _____

_____ Average

Nondestructive test results

MT _____ RT _____

Welding procedure

	Groove weld	Weld metal pad
Amperes	_____	_____
Arc travel speed (inches per minute)	_____	_____
No. passes, layer 1	_____	_____
No. passes, layer 2	_____	_____
No. passes, layer 3	_____	_____
No. passes, layer 4	_____	_____
No. passes, layer 5	_____	_____
No. passes, layer 6	_____	_____
No. passes, layer 7	_____	_____
No. passes, layer 8	_____	_____

Arc voltage _____
Amperes _____
Arc travel speed (inches per minute) _____

Electrode diameter

Maximum _____
Minimum _____

_____
Signature of responsible company official

_____
Date

Notes:

1. This form is applicable to the quality conformance tests required for each lot. One copy shall be forwarded to the consignee with each shipment.
2. See 4.5.

FIGURE 2. Certification of quality conformance tests. - Continued

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-24403/1D(SH)	Electrodes - Welding, Flux Cored, Ordinary Strength And Low Alloy Steel

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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